25901954 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

THIRD SEMESTER

Value Addition Course (VAC) - MG3VACCHE200 - FORENSIC CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions

Each question carries **1** marks

1. Corrosive alkali poisoning at autopsy shows: [K] / [CO1]

   a). White softened slippery mucosa b). Black mucosa

   c). Yellow fatty degeneration d). Blue gums

2. Which device is commonly used to detect metallic objects during security checks? [K] / [CO2]

   a). Ultrasound device b). Metal detector

   c). Infrared Scanner d). X- Ray scanner

3. UV rays are commonly used in forensic science for: [K] / [CO3]

   a). Detecting blood spatter patterns b). Identifying counterfeit documents

   c). . Taking fingerprints d). Analyzing DNA

4. Which spectroscopic technique is useful to detect the elemental composition of fibres [U] / [CO4]

a). FTIR b). Raman spectroscopy

c). UV-Vis spectroscopy d). X-ray fluorescence spectroscopy (XRF)

5. The antidote for Heparin overdose is [K] / [CO1]

a). Vitamin K b). Protamine sulphate

c). N-acetylcysteine d). Sodium thiosulfate

6. Which is the combustible material used in match box? [K] / [CO2]

a). Sulphur b). Potassium chloride

c). Calcium carbonate d). Sodium Nitrate

7. Which of the following is a common diamond simulant? [U] / [CO3]

a). Quartz b). Cubic zirconia

c). Topaz d). Garnet

8. Which plant is known as “suicide tree” due to its cardiac glycosides? [U] / [CO1]

a). Calotropis b). Cerbera odollam

c). Ricinus communis d). Abrus precatorius

9. The primary function of an incendiary bomb is: [K] / [CO2]

a). To produce sharpel b). To release toxic gas

c). To start fire d). To delay detonation

[1x9 = 9]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **4** marks

10. Explain how hair is used in biological samples in forensic detection of poison. [U] / [CO1]

11. Name two metals used in bullet cores. [K] / [CO2]

12. List out the side effects of corticosteroids. [K] / [CO4]

13. Explain how DNA differs from one person to another and how these differences are used in DNA fingerprinting. [U] / [CO4]

14. Give four postmortem features of hydrogen sulphide poisoning. [U] / [CO1]

15. Discuss about the military applications of RDX. [U] / [CO2]

16. Discuss about different types of glass fractures. [U] / [CO4]

[4x5 = 20]

Part C

Short Essay Questions

Answer any **3** questions

Each question carries **7** marks

17. Explain the methods and strategies for prevention and control of heavy metal contamination in seafood. [U] / [CO1]

18. Explain the forensic significance of identifying toxic versus corrosive substances in a crime investigation. [U] / [CO2]

19. (a)Discuss the forensic techniques of checking the silver line watermark in currency notes. (b) Explain the alloy analysis using Atomic Absorption Spectroscopy (AAS) for detecting counterfeit coins. [U] / [CO3]

20. What are the different types of forgery? [U] / [CO3]

21. Why is saliva considered a valuable biological fluid in forensic science, and how does it aid in criminal investigations? [K] / [CO4]

[7x3 = 21]